

MTE 203 – Advanced Calculus

Week 3 Tutorial Part 2 – Solutions

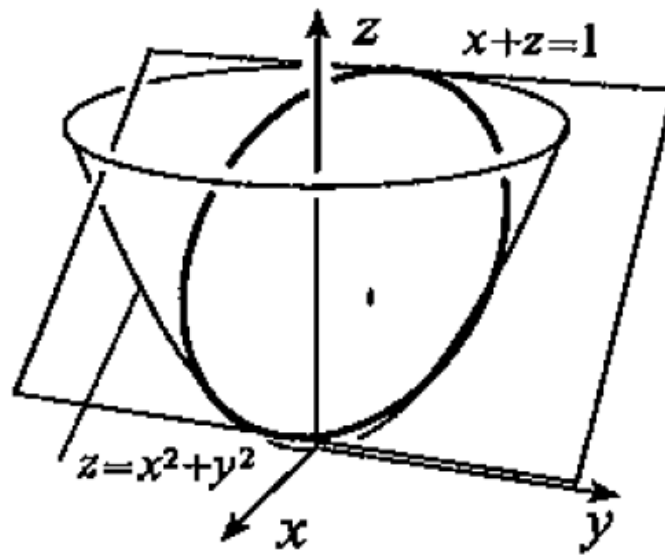
Problem 1:

A particle moves along the curve represented by the curve of intersection of the surfaces:

$$z = x^2 + y^2, \quad x + z = 1$$

- a. Find the equation for the projection of the curve in the x-y plane and draw the curve **[this part was solved during Lecture 4 – see LEARN]**

Solution:



The projection in the x-y plane can be obtained by equating both curves:

From the plane,

$$x + z = 1 \quad \rightarrow \quad z = 1 - x$$

Since,

$$z = x^2 + y^2 \quad \rightarrow \quad 1 - x = x^2 + y^2$$

re-arranging and factorizing,

$$x^2 + x - 1 + y^2 = 0$$

and,

$$\left(x + \frac{1}{2}\right)^2 + y^2 = \frac{5}{4}$$

Which is a circle centered at $\left(-\frac{1}{2}, 0\right)$ and radius equal to $\frac{\sqrt{5}}{2}$.

- b. Parametrize the curve of intersection so that it is directed counter-clockwise looking from above.

Using the following parametrization:

$$x = \frac{\sqrt{5}}{2} \cos t - \frac{1}{2}$$
$$\frac{\sqrt{5}}{2} \sin t$$

With $0 \leq t \leq 2\pi$, then,

$$z = 1 - x \rightarrow z = \frac{3}{2} - \frac{\sqrt{5}}{2} \cos t$$

And the vector function is given by,

$$\vec{r}(t) = \left(\frac{\sqrt{5}}{2} \cos t - \frac{1}{2}\right)\hat{i} + \left(\frac{\sqrt{5}}{2} \sin t\right)\hat{j} + \left(\frac{3}{2} - \frac{\sqrt{5}}{2} \cos t\right)\hat{k}$$

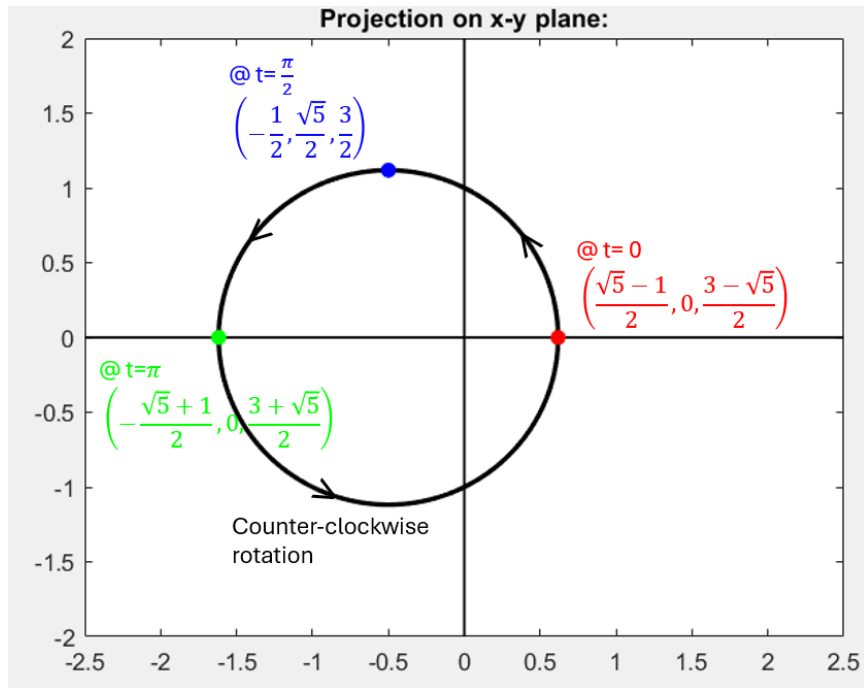
Checking the direction, we need three points:

$$t = 0 \rightarrow \left(\frac{\sqrt{5}-1}{2}, 0, \frac{3-\sqrt{5}}{2}\right)$$

$$t = \frac{\pi}{2} \rightarrow \left(-\frac{1}{2}, \frac{\sqrt{5}}{2}, \frac{3}{2}\right)$$

$$t = \pi \rightarrow \left(-\frac{\sqrt{5}+1}{2}, 0, \frac{3+\sqrt{5}}{2}\right)$$

Looking at these three points in the projection of the curve in the x-y plane, we can easily see that the curve is directed in the counter-clockwise direction, looking from above.



- c. Setup but do not evaluate the integral to find its length

The length of the curve is given by the following integral,

$$L = \int_{\alpha}^{\beta} |\vec{v}(t)| dt$$

Differentiating the vector function in part b, we can obtain the velocity vector as,

$$\vec{v}(t) = \frac{d\vec{r}(t)}{dt} = \left(-\frac{\sqrt{5}}{2} \sin t\right) \hat{i} + \left(\frac{\sqrt{5}}{2} \cos t\right) \hat{j} + \left(\frac{\sqrt{5}}{2} \sin t\right) \hat{k}$$

The speed is therefore given by,

$$|\vec{v}(t)| = \sqrt{\frac{5}{4} (\sin^2 t + \cos^2 t + \sin^2 t)}$$

Which can be re-written as,

$$|\vec{v}(t)| = \frac{\sqrt{5}}{2} \sqrt{1 + \sin^2 t}$$

And the length of the curve is given by,

$$L = \frac{\sqrt{5}}{2} \int_0^{2\pi} \sqrt{1 + \sin^2 t} dt$$

- d. Calculate the unit tangent $\hat{T}$ to the curve at the point $(-\frac{1}{2}, \frac{\sqrt{5}}{2}, \frac{3}{2})$

The unit tangent vector is given by,

$$\hat{T}(t) = \frac{\vec{v}(t)}{|\vec{v}(t)|} = \frac{-\sin t \hat{i} + \cos t \hat{j} + \sin t \hat{k}}{\sqrt{1 + \sin^2 t}}$$

At point $(-\frac{1}{2}, \frac{\sqrt{5}}{2}, \frac{3}{2}) \rightarrow t = \frac{\pi}{2}$, and the unit tangent vector is

$$\hat{T}\left(\frac{\pi}{2}\right) = \frac{-\hat{i} + \hat{k}}{\sqrt{2}}$$

$$\hat{T}\left(\frac{\pi}{2}\right) = \frac{1}{\sqrt{2}}(-1, 0, 1)$$

- e. Calculate the curvature at the point $(-\frac{1}{2}, \frac{\sqrt{5}}{2}, \frac{3}{2})$

The curvature of a curve parametrized by the parameter t is given by,

$$k = \frac{1}{|\vec{v}(t)|} \left| \frac{d\hat{T}}{dt} \right|$$

We can observe that although very doable, using the above formula for this specific problem will require differentiating a function with a square root in the denominator, which usually takes long.

In this case, a better formula will be to use the one derived in Lecture 10,

$$k = \frac{|\dot{\mathbf{r}} \times \ddot{\mathbf{r}}|}{|\dot{\mathbf{r}}|^3}$$

$$\begin{aligned}\dot{\mathbf{r}}(t) = \vec{v}(t) &= \left(-\frac{\sqrt{5}}{2} \sin t\right) \hat{i} + \left(\frac{\sqrt{5}}{2} \cos t\right) \hat{j} + \left(\frac{\sqrt{5}}{2} \sin t\right) \hat{k} \\ \ddot{\mathbf{r}}(t) = \frac{d\vec{v}(t)}{dt} &= \left(-\frac{\sqrt{5}}{2} \cos t\right) \hat{i} + \left(-\frac{\sqrt{5}}{2} \sin t\right) \hat{j} + \left(\frac{\sqrt{5}}{2} \cos t\right) \hat{k}\end{aligned}$$

$$\dot{\mathbf{r}} \times \ddot{\mathbf{r}} = \frac{5}{4} \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -\sin t & \cos t & \sin t \\ -\cos t & -\sin t & \cos t \end{vmatrix} = \frac{5}{4} (\sin^2 t + \cos^2 t) \hat{i} + \frac{5}{4} (\sin^2 t + \cos^2 t) \hat{k}$$

$$\dot{\mathbf{r}} \times \ddot{\mathbf{r}} = \frac{5}{4} (\hat{i} + \hat{k})$$

$$|\dot{\mathbf{r}} \times \ddot{\mathbf{r}}| = \frac{5\sqrt{2}}{4}$$

$$|\dot{\mathbf{r}}|^3 = \left(\frac{\sqrt{5}}{2} \sqrt{1 + \sin^2 t}\right)^3$$

$$|\dot{\mathbf{r}}|^3 = \frac{5\sqrt{5}}{8} (1 + \sin^2 t)^{\frac{3}{2}}$$

$$k = \frac{|\dot{\mathbf{r}} \times \ddot{\mathbf{r}}|}{|\dot{\mathbf{r}}|^3} = \frac{\frac{5\sqrt{2}}{4}}{\frac{5\sqrt{5}}{8} (1 + \sin^2 t)^{\frac{3}{2}}} = \frac{2\sqrt{10}}{5} (1 + \sin^2 t)^{-\frac{3}{2}}$$

$$k = \frac{2\sqrt{10}}{5} (1 + \sin^2 t)^{-\frac{3}{2}} \Big|_{P_0(t=\frac{\pi}{2})} = \frac{\sqrt{5}}{5}$$

For a length and difficulty comparison, try the alternate formula for curvature.

$$k = \frac{1}{|\vec{v}(t)|} \left| \frac{d\hat{T}}{dt} \right|$$

The speed was found in part c, and the unit tangent vector in part d.

$$\begin{aligned}|\vec{v}(t)| &= \frac{\sqrt{5}}{2} \sqrt{1 + \sin^2 t} \\ \hat{T}(t) &= \frac{-\sin t \hat{i} + \cos t \hat{j} + \sin t \hat{k}}{\sqrt{1 + \sin^2 t}}\end{aligned}$$

Differentiating the unit tangent vector, starting with the first component and applying the quotient rule.

$$\begin{aligned}
 \frac{d}{dt} \left(\frac{-\sin t}{\sqrt{1 + \sin^2 t}} \right) &= - \frac{\frac{d}{dt}(\sin t) \sqrt{1 + \sin^2 t} - \sin t \frac{d}{dt}(\sqrt{1 + \sin^2 t})}{\sqrt{1 + \sin^2 t}^2} \\
 &= - \frac{\cos t \sqrt{1 + \sin^2 t} - \sin t \left(\frac{1}{2} (1 + \sin^2 t)^{-\frac{1}{2}} (2 \sin t \cos t) \right)}{1 + \sin^2 t} \\
 &= - \frac{\cos t \sqrt{1 + \sin^2 t} - \frac{\cos t \sin^2 t}{\sqrt{1 + \sin^2 t}}}{1 + \sin^2 t} \\
 &= - \frac{\cos t}{\sqrt{1 + \sin^2 t}} + \frac{\cos t \sin^2 t}{(1 + \sin^2 t)^{\frac{3}{2}}} \\
 &= - \frac{\cos t (1 + \sin^2 t)}{(1 + \sin^2 t)^{\frac{3}{2}}} + \frac{\cos t \sin^2 t}{(1 + \sin^2 t)^{\frac{3}{2}}} \\
 &= \frac{-\cos t - \cos t \sin^2 t + \cos t \sin^2 t}{(1 + \sin^2 t)^{\frac{3}{2}}} \\
 &= \frac{-\cos t}{(1 + \sin^2 t)^{\frac{3}{2}}}
 \end{aligned}$$

Repeating this for each component, we obtain $\frac{d\hat{T}}{dt}$.

$$\frac{d\hat{T}}{dt} = \frac{-\cos t}{(1 + \sin^2 t)^{\frac{3}{2}}} \hat{i} - \frac{\sin^3 t + (1 + \cos^2 t) \sin t}{(1 + \sin^2 t)^{\frac{3}{2}}} \hat{j} + \frac{\cos t}{(1 + \sin^2 t)^{\frac{3}{2}}} \hat{k}$$

Substituting the point of interest for simplicity, and calculating the magnitude and the speed at the point gives everything needed to calculate the curvature.

$$\begin{aligned}
 \left. \frac{d\hat{T}}{dt} \right|_{P_0(t=\frac{\pi}{2})} &= \left(0, \frac{1}{\sqrt{2}}, 0 \right) \\
 \left| \left. \frac{d\hat{T}}{dt} \right| \right|_{P_0(t=\frac{\pi}{2})} &= \frac{1}{\sqrt{2}} \\
 \left| \vec{v} \left(\frac{\pi}{2} \right) \right| &= \frac{\sqrt{10}}{2}
 \end{aligned}$$

Finally, we calculate the curvature. For this problem, this solution approach is more tedious and lengthy due to the complex derivatives required.

$$k = \frac{1}{|\vec{v}(t)|} \left| \frac{d\hat{T}}{dt} \right|$$

$$k = \frac{\frac{1}{\sqrt{2}}}{\frac{\sqrt{10}}{2}}$$

$$k = \frac{\sqrt{5}}{5}$$

Problem 2 [S.11.12 Prob 26]:

- a. Calculate $\hat{T}, \hat{N}, \hat{B}$ for the curve given by the following parametric equations:

$$x = \cos t,$$

$$y = \sin t,$$

$$z = t.$$

- b. Express the vector $\vec{F} = x\hat{i} + xy^2\hat{j} + \hat{k}$ in terms of $\hat{T}, \hat{N}$ and $\hat{B}$.

Solution:

- a. For the given parametric equations, we can write the position vector as,

$$\vec{r} = \cos t \hat{i} + \sin t \hat{j} + t\hat{k}$$

Recall that,

$$\hat{T} = \frac{d\vec{r}}{dt} / \left| \frac{d\vec{r}}{dt} \right|$$

$$\hat{N} = \frac{d\hat{T}}{dt} / \left| \frac{d\hat{T}}{dt} \right|$$

$$\hat{B} = \hat{T} \times \hat{N}$$

Since:

$$\vec{r} = \cos t \hat{i} + \sin t \hat{j} + t\hat{k}$$

$$\frac{d\vec{r}}{dt} = \sqrt{\sin^2 t + \cos^2 t + 1} = -\sin t \hat{i} + \cos t \hat{j} + \hat{k}$$

And,

$$\left| \frac{d\vec{r}}{dt} \right| = \sqrt{2}$$

Hence,

$$\hat{T} = \frac{-\sin t \hat{i} + \cos t \hat{j} + \hat{k}}{\sqrt{2}},$$

$$\frac{d\hat{T}}{dt} = \frac{-\cos t \hat{i} - \sin t \hat{j}}{\sqrt{2}}$$

On the other hand,

$$\hat{N} = \frac{\frac{d\hat{T}}{dt}}{\left| \frac{d\hat{T}}{dt} \right|} = -\cos t \hat{i} - \sin t \hat{j}.$$

Finally,

$$\hat{B} = \hat{T} \times \hat{N} = \left(\frac{-\sin t \hat{i} + \cos t \hat{j} + \hat{k}}{\sqrt{2}} \right) \times (-\cos t \hat{i} - \sin t \hat{j})$$

$$\hat{B} = \frac{1}{\sqrt{2}} \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -\sin t & \cos t & 1 \\ -\cos t & -\sin t & 0 \end{vmatrix}$$

$$\hat{B} = (\sin t \hat{i} - \cos t \hat{j} + \hat{k})/\sqrt{2}$$

The components of $\vec{F}$ in the direction of $\hat{T}$, $\hat{N}$ and $\hat{B}$ are,

$$F_T = \vec{F} \cdot \hat{T} = (x, xy^2, 1) \cdot \frac{(-\sin t, \cos t, 1)}{\sqrt{2}} = \frac{1}{\sqrt{2}}(-x \sin t + xy^2 \cos t + 1)$$

$$F_T = \frac{1}{\sqrt{2}}(-\cos t \sin t + \sin^2 t \cos^2 t + 1)$$

$$F_N = \vec{F} \cdot \hat{N} = (x, xy^2, 1) \cdot (-\cos t, -\sin t, 0) = -(x \cos t + xy^2 \sin t)$$

$$F_N = -(\cos^2 t + \cos t \sin^3 t)$$

$$F_B = \vec{F} \cdot \hat{B} = (x, xy^2, 1) \cdot \frac{(\sin t, -\cos t, 1)}{\sqrt{2}} = \frac{1}{\sqrt{2}}(x \sin t - xy^2 \cos t + 1)$$

$$F_B = \frac{1}{\sqrt{2}}(\cos t \sin t - \sin^2 t \cos^2 t + 1)$$

We can therefore write $\vec{F}$ as,

$$\vec{F} = F_T \hat{T} + F_N \hat{N} + F_B \hat{B};$$

or

$$\begin{aligned} \vec{F} = & \frac{1}{\sqrt{2}}(-\cos t \sin t + \sin^2 t \cos^2 t + 1) \hat{T} - (\cos^2 t + \cos t \sin^3 t) \hat{N} + \\ & \frac{1}{\sqrt{2}}(\cos t \sin t - \sin^2 t \cos^2 t + 1) \hat{B} \end{aligned}$$